

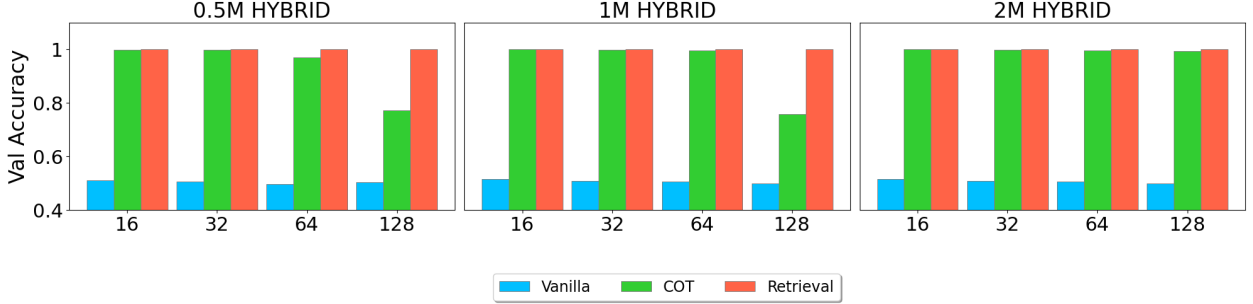


Figure 5: We train Mamba with one additional Transformer layer with a frozen word embedding and decoding head of three different model sizes (0.5M, 1M, 2M) on IsTree with three different sizes of graph (16, 32, 64) under three different setups. **Vanilla** means the model directly predicts the label. **COT** means the model will generate a chain-of-thought process based on DFS (see Algorithm 1) before prediction. **Retrieval** means the model will generate the chain of search queries and reasoning before prediction (see Algorithm 2).

6 Empirical Validation

We validate our theoretical findings through synthetic and natural language experiments: CoT alone cannot close the performance gap between RNNs and Transformers, but enhancing the in-context retrieval capability can narrow the gap.

6.1 Validation on Synthetic Task: IsTree

First, we validate our theoretical findings on the IsTree task introduced in Section 4.2.2. We generate a synthetic dataset for this task, where each data point is a tokenized graph concatenated with an answer of YES or NO, potentially with a CoT inserted in between. We then train RNNs and Transformers as language models on this dataset autoregressively.

Experiment Details. To generate the graph, we follow the procedure described in the proof of Theorem 4.7 (see Figure 3). The CoT data is generated using Algorithm 1 and the retrieval data is generated using Algorithm 2. For the CoT model, we decode the reasoning path during inference time until we reach the first YES or NO up to a max token limit greater than the length of all ground truth CoT. For the data points that the model fails to give a prediction, we assume the model gets it correct with 0.5 probability. For the retrieval task, we omit the explicit format of the regular expression and only ask the model to generate the vertices and special tokens in the regular expression to shorten the length of the input sequence. The reported accuracy is calculated over a validation set of 5000 samples using the last iteration of the model. We defer the experiment details to Appendix B. The results are shown in Figures 2 and 5.

Results with CoT. Without CoT, both the Transformers and RNNs model cannot learn to solve the IsTree problem. CoT improves the performance of both Transformers and RNNs but the RNNs’ performance degrades sharply as the graph size increases and the Transformers consistently outperforms RNNs in this case. This is consistent with our theory that CoT can improve the expressiveness of the RNN models but the expressiveness is still not enough to solve IsTree (see Theorems 4.1 and 4.7).

Results with In-Context RAG. In-Context RAG allows all the models to reach near-perfect accuracy. This is consistent with our theory that retrieval augmentation via regular expression can improve the expressiveness of the RNN models to solve algorithmic tasks (see Theorems 5.3 and 5.4). The hybrid model (a Mamba model with one additional Transformer layer on the top) shows performance on par with the Transformer, which is consistent with our theory (see Theorems 5.7 and 5.8).

6.2 Validation on Hotpot-QA

We further conduct experiments on open-source LLMs based on Transformer, Mamba, and a hybrid architecture to show that for tasks that require stronger in-context retrieval capability, RNNs suffer more performance degradation compared to Transformers.

Experiment Details. We use Phi-1.5 1.3B (Li et al., 2023a) as our Transformer model. We further use two Mamba and Transformer-Mamba hybrid models distilled from Phi-1.5 with approximately the same parameters’ sizes (Bick et al., 2024) as our RNN and hybrid models. To test our theory, we use the Hotpot-QA (Yang et al., 2018) dataset. In the validation set of Hotpot-QA, each question is accompanied by a set of different related paragraphs from Wikipedia. 2 of these paragraphs contain useful information to answer this question. The model needs to retrieve the correct paragraphs and reason based on these paragraphs to get the answer to the question.

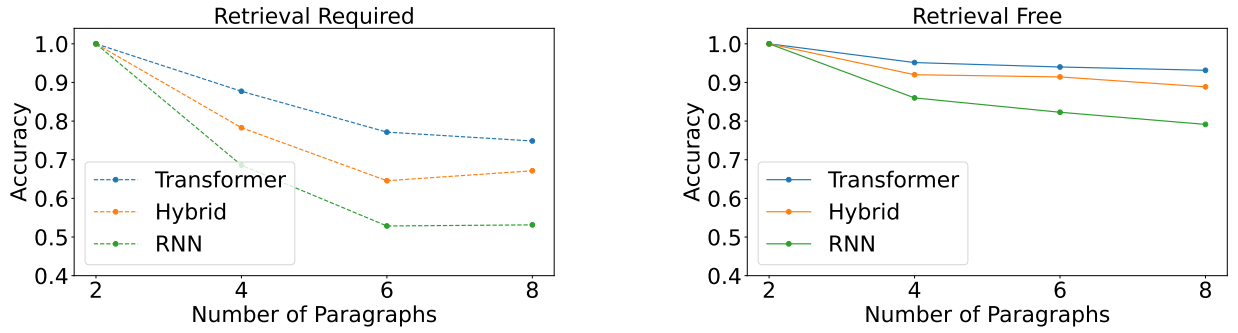


Figure 6: **Evaluation of Multi-Document Reasoning on Hotpot-QA.** n paragraphs are given to the model before the question and 2 among which contains required information to answer the question. (Left). When provided paragraphs are given in a random order, the task requires stronger in-context retrieval capabilities when n increases. RNNs’ performance drop more sharply, as predicted by theory. (Right). When the required paragraphs are always at the end of the provided paragraphs, the retrieval difficulty significantly decreases and all the models have stable performance with respect to n .

HotpotQA with Controlled Retrieval Difficulty. We consider the following version of Hotpot-QA with enhanced retrieval difficulty: We choose n paragraphs containing the correct paragraphs and order them randomly before the question. The model needs to answer the question based on the given paragraphs after CoT reasoning. This design allows us to use the hyperparameter n to control the difficulty of in-context retrieval in this task, with larger n corresponding to a higher difficulty.

Evaluation. We test our models under a 4-shot setting with Chain-of-Thought. The model we tested has varying performance even if $n = 2$. This is mostly due to their different capabilities to follow instructions and in-context examples. To mitigate this effect and highlight the impact of retrieval, we only test on a subset of 350 samples of the validation set where all the models can answer correctly given the correct paragraphs. This ensures perfect accuracy when $n = 2$ in both settings. We vary n in $\{2, 4, 6, 8\}$ and the result is shown in Figure 6.

RNNs’ performance drops sharply with increased retrieval difficulty. While all the model’s performance drops with increased retrieval difficulty (Figure 6, left), the RNN model has the largest drops. The hybrid model with only 4 attention layers performs significantly better than the RNN model. This validates our theory that RNN architectures’ limited in-context retrieval capabilities will impact their performance in reasoning and hybrid architecture is a potential solution to this limitation.

Deconfounding the effect of context length. The context length also increases with the number of provided paragraphs and can be a potential confounder. We then experiment with a controlled group: after choosing the same set of n paragraphs, we always order the correct paragraph at the end, which significantly simplifies the in-context retrieval process. In this case, all the models have stable performance when the number of paragraphs increases.

7 Conclusion and Discussion

This paper studies the representation gap between RNNs and Transformers on algorithmic problems. It is proved that CoT can improve RNNs but is insufficient to close the gap with Transformers: the inability of RNNs to perform in-context retrieval is a key bottleneck. To address this, we show that adopting In-Context RAG or appending a single Transformer layer to RNNs can enhance their in-context retrieval capabilities and close the representation gap with Transformers.

One limitation of this work is that the solution of In-Context RAG through regular expression is for the purpose of understanding the bottleneck of the representation power of RNNs, and may not be a practical method beyond IsTree since there is no existing training data for this type of RAG. Effectively enabling or eliciting LLMs’ capability to perform in-context RAG or other types of RAG is an interesting direction for future work. Second, appending a single Transformer layer to RNNs is a minimal example of making architectural changes to RNNs to improve their representation power while marginally increasing the memory cost. It is left unexplored what other architectural changes can pose a similar effect or enjoy a better trade-off between representation power and memory efficiency. Finally, we only study the aspect of representation power, and do not analyze the training dynamics and generalization of RNNs and Transformers. We believe this is the most challenging but important direction for future work.

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